BEE 4750/5750

Mini-Project 1: Dissolved Oxygen with Multiple Effluents

2026-09-23

Due Date
Thursday, 10/22/26, 9:00pm

Overview

Three facilities discharge into the same river. Additionally, there is a legacy sludge deposit on the riverbed. You will build a model of dissolved oxygen (DO) along the river, check whether four candidate treatment plans meet the state standard, test how reliable these plans are when one facility's effluent varies from day to day, and recommend a plan.

Instructions

- Students in BEE 4750 may work in groups of 2; students in BEE 5750 work individually.
- Submit a report as a PDF to Gradescope. Make sure to tag pages corresponding to a problem in Gradescope.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using Random
using Statistics
using Distributions
using Plots
using LaTeXStrings
```

The System

The river flows at $U = 6$ km/d. The reaeration rate is $k_a = 0.55\ \mathrm{d}^{-1}$; CBOD and NBOD decay at $k_c = 0.35\ \mathrm{d}^{-1}$ and $k_n = 0.25\ \mathrm{d}^{-1}$. Saturated DO is $C_s = 10$ mg/L. Upstream of the first facility the river carries 80,000 m³/d at 6.8 mg/L DO, 4.0 mg/L CBOD, and 3.0 mg/L NBOD.

Table 1: Effluent characteristics before treatment.

	Facility A	Facility B	Facility C
Type	municipal	municipal	food processing (private)
Location	0 km	15 km	32 km
Flow (m³/d)	22,000	10,000	18,000
DO (mg/L)	2.0	2.5	1.5
CBOD (mg/L)	66	54	80
NBOD (mg/L)	38	30	60

Decades of discharge before the current permits left a bed of settled organic solids between 20 km and 30 km downstream of Facility A. It exerts a benthic oxygen demand that is strongest at its upstream edge and tapers to nothing at its downstream edge:

$$S_B(x) = 1.6\left(1 - \frac{x - 20}{10}\right) \text{ mg /(L · d)} \qquad \text{for } \; 20 \le x \le 30,$$

and $S_B(x) = 0$ everywhere else.

The relevant regulatory standard from New York State requires DO to stay at or above **4 mg/L** at every point in the river.

As treatment options, each facility can install one level of treatment, which removes the given fraction of both CBOD and NBOD from its effluent. Costs are annualized, in thousands of dollars per year.

Table 2: Treatment levels and annualized costs ($1,000/yr).

Level	Removal	Facility A	Facility B	Facility C
None	0%	0	0	0
Primary	35%	310	180	265
Secondary	65%	720	395	640
Tertiary	85%	1,450	810	1,980

The regional water authority is considering four plans:

Table 3: Candidate treatment plans.

Plan	Facility A	Facility B	Facility C	Annual cost
1	secondary	primary	primary	$1,165,000
2	secondary	secondary	primary	$1,380,000
3	secondary	primary	secondary	$1,540,000
4	tertiary	primary	primary	$1,895,000

Here is this information summarized for you:

```
river_params = let
    ka, kc, kn = 0.55, 0.35, 0.25   # reaeration, CBOD decay, NBOD decay   [1/d]
    Cs, U = 10, 6                   # saturation DO [mg/L], velocity [km/d]
```

```
    (; ka, kc, kn, Cs, U)
end

# Conditions in the river upstream of Facility A
river_flow = 80_000     # [m³/d]
river_DO   = 6.8        # [mg/L]
river_CBOD = 4          # [mg/L]
river_NBOD = 3          # [mg/L]

# One entry per facility, in order going downstream: A, B, C
facility_location = [0, 15, 32]            # [km]
facility_flow     = [22_000, 10_000, 18_000]  # [m³/d]
facility_DO       = [2, 2.5, 1.5]          # [mg/L]
# Float64 arrays, so a Monte Carlo draw can be stored in them
facility_CBOD     = Float64[66, 54, 80]    # [mg/L], before treatment
facility_NBOD     = Float64[38, 30, 60]    # [mg/L], before treatment

# Benthic oxygen demand from the sludge bed [mg/(L·d)]
function sludge_demand(x)
    if 20 <= x <= 30
        return 1.6 * (1 - (x - 20) / 10)
    else
        return 0
    end
end

# Treatment levels, in order: none, primary, secondary, tertiary
treatment_removal = [0, 0.35, 0.65, 0.85]
# Annualized cost [$1,000/yr]; one row per facility (A, B, C), one column per level
treatment_cost = [0 310 720 1450;
                  0 180 395 810;
                  0 265 640 1980]

# Fraction of CBOD and NBOD removed at facilities A, B, C under each candidate plan
plan_removal = [
    [0.65, 0.35, 0.35],   # Plan 1
    [0.65, 0.65, 0.35],   # Plan 2
    [0.65, 0.35, 0.65],   # Plan 3
    [0.85, 0.35, 0.35],   # Plan 4
]
plan_cost = [1.165e6, 1.380e6, 1.540e6, 1.895e6]   # [$/yr]
```

Parts (Total: 100 Points)

Part 1 (15)

Model DO, CBOD, and NBOD as functions of distance from Facility A to 70 km downstream.

Part 1.1 (4)

Explain why you cannot used the closed-form Streeter-Phelps solution to model this system.

Part 1.2 (4)
Write the forward Euler update rules for DO, CBOD, and NBOD in terms of distance, including the sludge bed. Mix each effluent into the river at the grid point at the facility's location; choose step sizes that put a grid point exactly at 15 km and 32 km.

Part 1.3 (7)
Implement your model and plot DO against distance with **no treatment** at any facility. Mark the standard on the plot. Where is DO lowest, how low does it get, and why does the lowest point fall there rather than just after an effluent?

Part 2 (15)

Part 2.1 (9)
Run a convergence study on the **minimum DO** with no treatment. Build a reference solution at a very fine step, then test a sequence of successively halved step sizes against it. Report a table of step sizes, minimum DO, error against the reference, and the improvement between resolution sizes, then plot error against step size on log-log axes.

Part 2.2 (6)
State the step size you will use for the rest of the project and justify it. Does the slope match what we expect for forward Euler?

Part 3 (20)

Part 3.1 (8)
Evaluate the four plans in Table 3. For each, report the minimum DO, where it occurs, and whether the plan complies with the standard. Plot the four DO profiles on the same set of axes with the standard marked with a dashed red line.

Part 3.2 (6)
Plan 4 is the most expensive of the four, yet it leaves less margin above the standard than Plan 3. Explain why.

Part 3.3 (6)
The authority could instead dredge the sludge bed, which corresponds to removing it from your model. How much does the minimum DO change with no treatment, and how much of the original violation does that account for? Would this change the effectiveness of any of the plans?

Part 4 (30)
Suppose Facility C's waste stream varies from day to day: its dissolved oxygen, untreated CBOD, and untreated NBOD all change together. Assume they are independent, with

Facility C effluent	Distribution (mg/L)
DO	$\mathcal{N}(1.5, 0.5^2)$, at least 0
CBOD, before treatment	$\mathcal{N}(80, 12^2)$
NBOD, before treatment	$\mathcal{N}(60, 9^2)$

and everything else as before. Each Monte Carlo sample should draw all three.

Part 4.1 (8)
For each plan that complied in Part 3, and for Plan 1 with the sludge bed dredged, estimate the probability that the river violates the standard on a randomly chosen day. Report each estimate with a 95% confidence

interval, your sample size, and your seed. If a plan produces no violations in your sample, say what that does and does not tell you about its probability of violating. Finally, is it reasonable to draw Facility C's three quantities independently? If not, which way would that push your estimates?

Part 4.2 (8)

Your estimate for Plan 2 involves two sources of error: one from the discretization step size Δx, one from the Monte Carlo sample size n. They can be compared by examining their influence on the same modeled quantity of interest. Let's use the violation probability.

Hold n and the random seed fixed and vary Δx; then hold Δx fixed and vary n. Report both as a table. Which error is larger at the settings you used in 4.1, and, if you could spend some time reducing one, which would be the most useful?

Part 4.3 (4)

Your confidence interval reports one of those two errors and is silent about the other. Which one, and what might a reader who just encounters your interval wrongly conclude?

Part 4.4 (10)

The four candidates are only 4 of the 64 ways to assign treatment levels to the three facilities (Table 2). Find the cheapest plan whose probability of violating the standard is below 5%, without running Monte Carlo on all 64. First **screen** every plan with your deterministic model at average conditions, with each of Facility C's three quantities at its mean; then **sample**, running Monte Carlo only on plans that pass the screen, starting from the cheapest.

Report how many plans pass the screen, which plans you sampled, and the plan you find. Then explain why the screen is safe here: could a plan that fails it still have a violation probability below 5%?

Part 5 (20)

You are the engineer retained by the regional water authority, which must select one plan and defend it publicly. Facilities A and B are municipal and will pass their costs on to ratepayers; Facility C is a private food processor that will bear its own.

Part 5.1 (8)

Recommend one plan. Justify the choice, and explain how you weighed its cost against its probability of violating the standard.

Part 5.2 (7)

The authority's board includes members who prioritize ratepayer cost, members who prioritize river condition, and members concerned with how the burden falls between municipal and private dischargers. Which plan would each group favor, and why? Does your recommendation change under any of these priorities?

Part 5.3 (5)

What information, not available to you here, would most change your recommendation?

References

List any external references consulted, including classmates.